Lecture 11: Introduction to Entropy and the Second Law

CBE 20260 / Thermodynamics I

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Table of contents

1 The Arrow of Time

In the previous lecture, we established that the First Law of Thermodynamics is a conservation law. It tells us that energy cannot be created or destroyed, only transformed. However, the First Law fails to explain a fundamental aspect of nature: **Directionality**.

Processes in nature occur spontaneously in one direction but never in the reverse, even though the reverse process would satisfy the First Law. Let's look at three specific examples.

1.1 Case 1: The Gas Expansion

Imagine a rigid, insulated tank containing high-pressure gas ($P_{gas} > P_{atm}$). We open a valve, and the gas rushes out into the atmosphere.

- **Forward Process:** Gas flows from high pressure to low pressure.
- **Reverse Process:** Air from the room spontaneously rushes back into the tank, re-pressurizing it.
- **First Law Analysis:** If the tank is adiabatic ($Q = 0$) and rigid, energy is conserved in both directions. The First Law does not forbid the air from rushing back in, yet we never see it happen.

1.2 Case 2: Thermal Equilibrium

Consider two blocks, Block A (Hot) and Block B (Cold), brought into contact.

- **Forward Process:** Heat flows from A to B until $T_A = T_B$.
- **Reverse Process:** Two blocks at the same temperature spontaneously separate into a hot block and a cold block.
- **First Law Analysis:** As long as the heat lost by one equals the heat gained by the other ($Q_{hot} = -Q_{cold}$), energy is conserved. The First Law permits the "Magic Refrigerator," but nature forbids it.

1.3 Case 3: Mixing

Consider a box separated by a partition. On the left are red molecules; on the right are blue molecules. We remove the partition.

- **Forward Process:** The gases mix until they are uniformly distributed.
- **Reverse Process:** The mixed gases spontaneously separate back into pure red and pure blue sides.
- **First Law Analysis:** Mixing is an energy-neutral process for ideal gases. There is no energy barrier preventing un-mixing, yet it never occurs.

2 The Second Law of Thermodynamics

To account for this "directionality," we postulate the existence of a new property. This property must measure the "quality" of energy or the "disorder" of a system. We call this property **Entropy** (S).

We can now state the **Second Law of Thermodynamics**:

! The Second Law

There exists a property called entropy (S), which is an intrinsic, extensive property of a system. For an **isolated system**, the change in entropy is always positive or zero.

$$\Delta S_{isolated} \geq 0$$

- $\Delta S > 0$**:** The process is **Irreversible** (Real, Spontaneous).
- $\Delta S = 0$**:** The process is **Reversible** (Ideal, Equilibrium).
- $\Delta S < 0$**:** The process is **Impossible**.

Mathematically, since entropy is increasing for an isolated system, we say

$$\frac{dS}{dt} > 0$$

In this way, entropy is like time: it always moves forward. The "arrow of time" is the "arrow of increasing entropy." Unlike the energy "conservation" of the First Law, the Second Law is a "non-decreasing" law. It tells us that entropy can only stay the same or increase, never decrease. Thus we can have entropy *generation*

$$\frac{dS}{dt} = \dot{S}_{gen} \geq 0$$

Entropy is maximized at equilibrium, so we can also say that for an isolated system at equilibrium:

$$\frac{dS}{dt} = 0$$

3 Defining Entropy Mathematically

How do we calculate this property? For a closed system undergoing a process, the differential change in entropy (dS) is defined related to the heat transfer occurring in a **reversible** path:

$$dS = \frac{dQ_{rev}}{T} \tag{1}$$

Where:

- dS is the change in entropy of the system.
- dQ_{rev} is the heat transfer entering the system if the process were carried out reversibly.
- T is the absolute temperature (e.g. Kelvin) at the boundary where heat transfer occurs.

Note that the definition of dS is based on a **reversible** process, even if the actual process is irreversible. This is a crucial point that allows us to calculate entropy changes for any process by imagining a reversible path between the initial and final states. Also, heat fluxes across system boundaries are the only way to change the entropy of a system. Work does not directly change entropy, but it can indirectly affect it by changing the state of the system.

3.1 Entropy as a State Function

Crucially, S is a **State Function**.

$$\Delta S = S_2 - S_1 = \int_1^2 \frac{dQ_{rev}}{T} \tag{2}$$

Even if the *actual* process is irreversible (like the high pressure gas leaving the tank), we calculate ΔS by imagining a reversible path connecting State 1 and State 2. (We will shortly see that we don't need to construct imaginary reversible paths in many cases; we can calculate ΔS directly from state properties.)

3.2 Entropy Generation (S_{gen})

For any general process (not necessarily reversible), we write the entropy balance for a closed system as:

$$dS = \frac{dQ}{T} + dS_{gen}$$

or in rate form:

$$\frac{dS}{dt} = \frac{\dot{Q}}{T} + \dot{S}_{gen}$$

Note that it is possible to have multiple heat flows at different temperatures, in which case there would be multiple $\frac{Q}{T}$ terms, one for each heat flow. The key point is that the total change in entropy of the system is the sum of two contributions:

- $\frac{Q}{T}$: Entropy transferred accompanying heat flow.
- S_{gen}: Entropy **generated** inside the system due to irreversibilities (friction, mixing, finite temperature gradients).

The Second Law requirement is specifically:

$$S_{gen} \geq 0$$

Obviously, for a reversible process

$$dS = \frac{dQ_{rev}}{T}$$

which is Eq. 1 again.

4 Proving Directionality: The Two Blocks

Let's apply this mathematical definition to **Case 2** (The Hot and Cold Blocks) to prove that the Second Law correctly predicts the direction of heat flow.

System: Two rigid blocks, A and B, isolated from the surroundings.

- Block A is at temperature T_A.
- Block B is at temperature T_B.
- A small amount of heat, $|Q|$, transfers from one to the other.

Since the total system is isolated (adiabatic), $\Delta S_{surr} = 0$. Therefore, $\Delta S_{total} = \Delta S_A + \Delta S_B$.

Using our definition $dS = dQ/T$:

$$\Delta S_A = \frac{Q_A}{T_A} \quad \text{and} \quad \Delta S_B = \frac{Q_B}{T_B}$$

From the First Law ($Q_A + Q_B = 0$), let's define heat flow direction.

4.1 Scenario 1: Heat flows from Hot (T_A) to Cold (T_B)

Let $T_A > T_B$. Heat leaves A ($Q_A = -Q$) and enters B ($Q_B = +Q$).

$$\Delta S_{total} = \frac{-Q}{T_A} + \frac{Q}{T_B}$$

$$\Delta S_{total} = Q\left(\frac{1}{T_B} - \frac{1}{T_A}\right)$$

Since $T_A > T_B$, the term $(1/T_B)$ is larger than $(1/T_A)$. The term in the parentheses is **positive**.

$$\Delta S_{total} > 0$$

Conclusion: $S_{gen} > 0$. The process is **Possible** and Irreversible.

4.2 Scenario 2: Heat flows from Cold (T_B) to Hot (T_A)

Let $T_A > T_B$. Assume heat leaves B ($Q_B = -Q$) and enters A ($Q_A = +Q$).

$$\Delta S_{total} = \frac{Q}{T_A} - \frac{Q}{T_B}$$

$$\Delta S_{total} = Q\left(\frac{1}{T_A} - \frac{1}{T_B}\right)$$

Since $T_A > T_B$, $(1/T_A) < (1/T_B)$. The term in parentheses is **negative**.

$$\Delta S_{total} < 0$$

Conclusion: $S_{gen} < 0$. The process is **Impossible**.

The math confirms our intuition: Heat flows spontaneously only from hot to cold.

5 Driving Forces and Entropy Generation

We stated earlier that $\Delta S = 0$ for a reversible process. A reversible process is one where the "driving forces" (differences in T, P, etc.) are infinitesimal.

Let's look at the entropy generation equation for the blocks again:

$$\dot{S}_{gen} = \dot{Q}\left(\frac{1}{T_{cold}} - \frac{1}{T_{hot}}\right) = \dot{Q}\left(\frac{T_{hot} - T_{cold}}{T_{hot}T_{cold}}\right)$$

Let's assume the heat transfer rate is proportional to the temperature difference (Newton's Law of Cooling): $\dot{Q} = k(T_{hot} - T_{cold})$.

Substitute this into the expression:

$$\dot{S}_{gen} = k(T_{hot} - T_{cold})\left(\frac{T_{hot} - T_{cold}}{T_{hot}T_{cold}}\right)$$

$$\dot{S}_{gen} = \frac{k(T_{hot} - T_{cold})^2}{T_{hot}T_{cold}}$$

If we define the driving force as $\Delta T = T_{hot} - T_{cold}$:

$$\dot{S}_{gen} \propto (\Delta T)^2$$

6 The Takeaway

Entropy generation (which represents lost work or inefficiency) scales with the **square** of the driving force.

1. **Large gradients** (large ΔT) $\rightarrow$ High Entropy Generation $\rightarrow$ Highly Irreversible.
2. **Small gradients** (small ΔT) $\rightarrow$ Low Entropy Generation $\rightarrow$ Approaches Reversibility.

This explains why, in our "Grains of Sand" piston example, we needed to make changes in small increments to keep the process reversible. Nature penalizes "rushing" (large driving forces) with large entropy generation.